NAVYANTRA – BUILD BEYOND BOUNDARIES

24-Hour National-Level Hardware Hackathon

ABSTRACT SUBMISSION TEMPLATE

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Team Name: InnoVolt

Project Title: EDGEGUARD V1 – Intelligent Edge Monitoring & Protection System

Domain: Smart Infrastructure and Industrial Automation

1. PROBLEM STATEMENT

Machine failures rarely begin as sudden events; they often leave measurable signs such as rising temperature, abnormal current, voltage variation, and mechanical vibration. However, these signals are commonly monitored independently, making it difficult to understand the machine's overall condition before a serious fault develops. EDGEGUARD addresses this gap by exploring a compact, low-cost system that transforms multiple physical indicators into a unified risk assessment and enables timely protective intervention.

2. PROPOSED SOLUTION

EDGEGUARD V1 is a real-time, edge-intelligent monitoring and protection system that treats a machine as a combination of interacting physical conditions rather than isolated sensor readings. An ESP32 acquires motor surface temperature, vibration, current, and voltage data through an LM35, MPU6050, and INA219 respectively. Instead of merely displaying these parameters, the system converts them into individual risk indicators and combines them into a machine-condition risk score. The controller classifies the operating state as NORMAL, WARNING, or CRITICAL. As risk increases, the system provides an early warning through visual and audible indicators. At a critical condition, the system goes beyond monitoring and performs a physical protection action by disconnecting the motor through a relay. This creates a closed loop from physical sensing to intelligent decision-making and protective intervention at the edge.

3. METHODOLOGY / WORKING PRINCIPLE

EDGEGUARD V1 operates on the principle **Sense** → **Analyze** → **Decide** → **Protect**.

The system continuously observes the physical behaviour of a 12V DC motor through three complementary sensing dimensions. An LM35 is physically attached to the motor's metal body to capture surface-temperature rise associated with motor heating. An MPU6050/GY-521 captures mechanical vibration, providing information about abnormal mechanical behaviour. An INA219 monitors the electrical side by measuring motor-supply voltage and current.

The ESP32 acts as the edge intelligence layer. Instead of forwarding raw measurements for external processing, it acquires and processes the sensor information locally. Basic filtering is applied to reduce unwanted fluctuations, after which each parameter is mapped to a normalized risk value according to predefined operating limits. These individual indicators are then combined to generate an overall machine-condition risk score.

The decision engine continuously evaluates this score and places the system into one of three states: NORMAL, WARNING, or CRITICAL. NORMAL indicates stable operation. WARNING represents a developing abnormal condition and triggers an alert. CRITICAL indicates that the combined risk has reached a predefined protection region.

The prototype can demonstrate changing electrical and mechanical behaviour through controlled motor loading, while controlled motor heating can demonstrate surface-temperature variation at the LM35. These experiments allow the system to observe multiple forms of abnormal behaviour without intentionally damaging or stalling the motor.

When a critical condition is identified, the ESP32 activates a relay-based protection mechanism to disconnect the motor supply. LEDs and a buzzer provide immediate local feedback.

Thus, EDGEGUARD demonstrates more than condition monitoring: it creates a closed-loop path from physical evidence to risk assessment, decision-making, warning, and physical intervention.

4. KEY COMPONENTS / TECHNOLOGIES

The core of EDGEGUARD V1 is an **ESP32 DevKit** implementing local edge processing and the protection decision engine. An **LM35** is physically coupled to the motor body for surface-temperature monitoring, while an **MPU6050/GY-521** captures mechanical vibration. An **INA219** provides electrical voltage and current measurements. A **12V DC motor** represents the monitored machine, with an **L298N motor driver** enabling controlled operation during experiments. A **5V relay module** provides the final physical cutoff mechanism. LEDs and a buzzer provide immediate system-state and fault alerts. The firmware uses the Arduino framework and PlatformIO, while Wokwi supports initial ESP32 simulation and validation. The architecture is intentionally modular for future expansion.

5. EXPECTED OUTCOME

The expected outcome is a working proof-of-concept demonstrating how multiple physical symptoms of machine stress can be transformed into a real-time protection decision. EDGEGUARD will detect changes in motor surface temperature, current, voltage, and vibration; process these signals locally; and represent machine condition through NORMAL, WARNING, and CRITICAL states. Controlled loading and heating experiments will demonstrate corresponding changes in monitored parameters. When the combined risk reaches the critical region, the system will trigger an alert and physically disconnect the motor through the protection relay. The prototype aims to demonstrate the practical transition from **“detecting a problem” to “acting on the problem”**, providing a foundation for scalable, low-cost machine-condition monitoring, preventive protection, and future intelligent industrial safety applications in diverse industrial environments.

6. INNOVATION / UNIQUE FEATURE

The defining feature of EDGEGUARD V1 is its shift from parameter monitoring to condition-aware protection. Rather than treating temperature, electrical behaviour, and vibration as isolated alarms, the system combines their risk indicators into a unified machine-condition assessment. This creates a more meaningful picture of developing abnormal behaviour and enables graduated responses through NORMAL, WARNING, and CRITICAL states. Its second key feature is the direct connection between intelligence and action: the system does not stop at displaying a fault; a critical decision can physically disconnect the load. The LM35’s direct coupling to the motor body strengthens the thermal-monitoring approach, while the modular architecture allows future integration of additional sensing, data logging, anomaly detection, and hardware-based processing. EDGEGUARD is therefore designed as a foundation for turning machine data into timely protective action.